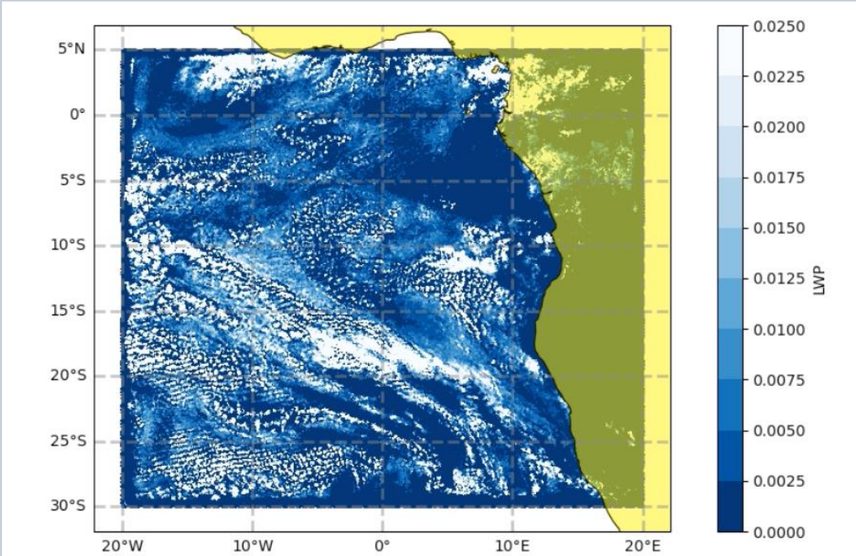
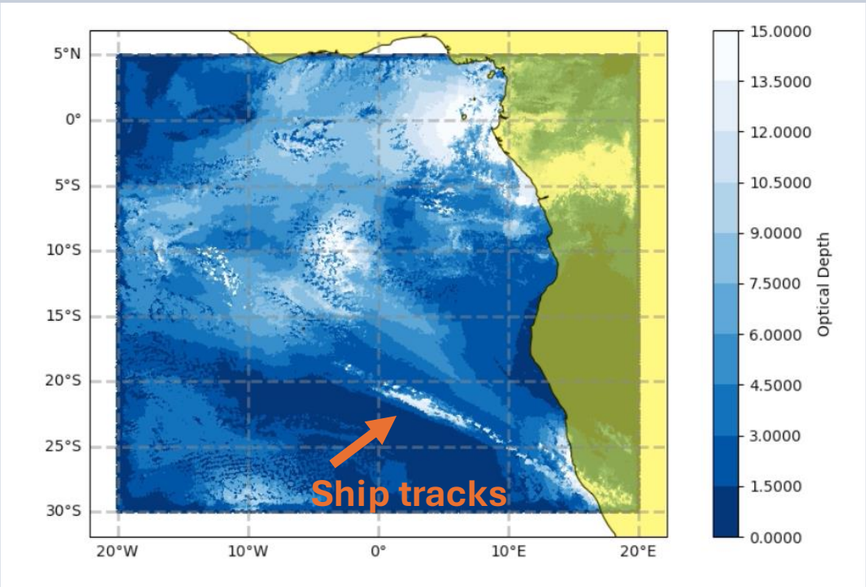
New Topics, Tools, and Terrain: Aerosol Stratocumulus Interactions

Present: Understanding the Sensitivity of Marine Stratocumulus Clouds to Sea Salt Perturbations

5 km Resolution ICON-HAM-lite 1-Sept-2018 9:00:00



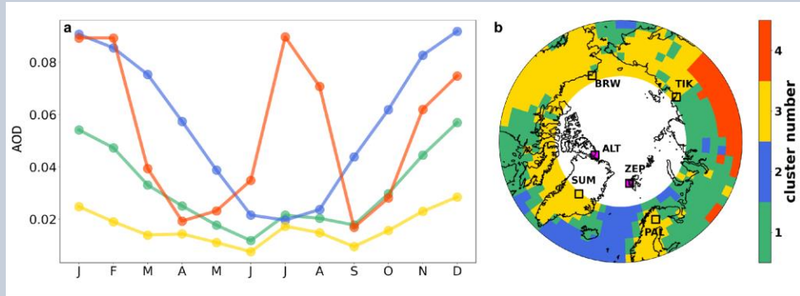


1. Using the ICON-HAM-lite model, we find that some mesoscale convective organization is evident in 5km model runs, as are brightening effects of shipping emissions on clouds.

2. Next, we will examine the sensitivity of cloud organization to higher resolutions (1-2.5 km).

3. We will then test the sensitivity of cloud properties to different sea spray emissions parameterizations, leveraging source function diversity and implementing sea spray injectors into the HAM-lite model.

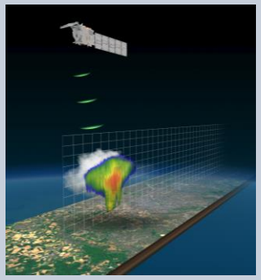
Past: The (Many) Processes Governing Aerosol Spatiotemporal Variability



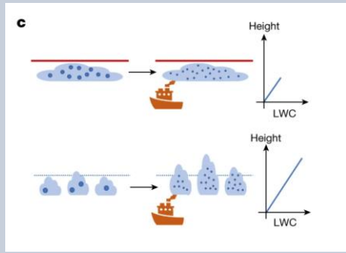
Using satellite data and K-means clustering, we found that the Arctic shows four distinct subregions of aerosol seasonality, each with their own seasonal drivers.

Also ask me about boreal wildfires, aerosols during covid, and aerosol remote sensing!

Future: Convective Invigoration in Ship Tracks



ESA – EarthCARE Dopper



"Invisible Ship Tracks" Manshausen et al., 2022

Using EarthCARE and ship track data, we will assess the extent to which ship tracks deepen (rather than just brighten) clouds under different meteorological conditions.

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